[TEGMARK] PRE-REGISTRATION (before the number) — Casey's "if it comes back 4, the subdomain forces 3"

Grace | 2026-07-27 Mon | Casey's steer: if the generation count derives to 4, the fermion-carrying SUBDOMAIN (probably SO(5,2)-related) forces 3. This is a real structural instinct AND it lands on the day's most target-sensitive spot — so I pre-register it now, before the count is read, with the target-innocence condition explicit. Pre-registering is exactly what turns it from a post-hoc rescue into a legitimate contingency (the same discipline that caught the smuggled "three" in four places today).

> ? UPDATED 2026-07-27 (K957) — where this note says the E_0 ambiguity "may favor 4": superseded. Cal read FG-2014 Table 2 — the SO(5,2) spinor singleton ground state is E_0=2 (the internal value was right; the "5/2" was a category error — naive n_C/2, not the spinor's conformal weight). So the honest lean is back to 3, and the count is the reduction structure (where the module develops a submodule), NOT a norm-signature (the singleton norm is vacuous — an infinite positive tower). This contingency (IF the reduction returns 4) still stands — the reduction computation hasn't run — but it no longer rides on the dead 5/2. Also (Cal): "n−1=4" is the emergent SPACETIME dimension (Selector-2), NOT a generation-count formula — do not import it as the count.

The instinct, stated fairly

The naive full-domain scalar singleton tower carries the E_0/threshold ambiguity (3-vs-4). But the physical fermions may not be counted on the full tower — they may live on a subdomain / sub-representation, whose OWN count is the physical answer. If that sub-object forces 3, then 3 is right even if the full tower admits a 4th rung.

⋆ The condition that decides insight vs. rescue (the whole point)

This is target-innocent if and only if the fermion sub-locus is forced independently of, and prior to, the generation count — named before the number, for reasons that don't mention "3." It is target-aware (the exact trap we fenced all day) if we reach for "some subdomain gives 3" *after* seeing 4.

Why the instinct has a genuinely forced candidate (this is the encouraging part)

There ARE fermion sub-loci in the corpus that are forced *independent of the count*: - The chargeless S^4 locus — the ν_R condensate demonstrably sits there (F571/F582), forced by its electric neutrality ($Q\langle\varphi\rangle=0$, F583), NOT by generations. - The SO(4,2) conformal boundary sector — the EM/spacetime sector (F66, Casey #14), unbroken at every scale, forced by masslessness of the photon and the conformal descent — again nothing to do with the count. So "the generations are counted on the forced fermion sub-locus, not the full scalar tower" is a hypothesis with an *independently-sourced* subject. That is what makes Casey's move disciplined, not a rescue.

The pre-registered test (commit now)

1. Run the threshold+signature count on BOTH the full D_IV^5 scalar tower AND the forced fermion sub-locus (S^4 / SO(4,2) conformal sector — whichever is the *derived* fermion carrier), by the identical rule, before knowing either answer.
2. Sub-locus count is the physical answer ONLY IF the sub-locus is the *forced* fermion carrier (established by F571/F582/F66, not by the count) AND its count comes out by its own mechanism (its own threshold), not because we selected the sub-locus that yields 3.
3. If both full-domain AND the forced sub-locus give 4 → four generations is the honest answer, we publish it. The sub-locus does NOT get to override a genuine 4.
4. The E7 check runs identically on E7's corresponding sub-locus — the exclusion must survive the same sub-locus rule, or it's not an exclusion.

Honest read (not deference — the discipline both ways)

- Casey's instinct is structurally sound: generation-counts on symmetric domains often live on a sub-representation, not the full tower. Worth building.
- But I hold the knife: adopting it *because* it yields 3 would be the fourth-place smuggle of the day. The guard is that the sub-locus (S^4/SO(4,2)) is independently forced — which it is — and that its count still comes out by mechanism.
- And the K955 payoff stands regardless: color forces the domain independent of the count. So whether the honest answer is 3 (full or sub-locus) or 4, D_IV^5 is unchanged. The subdomain hypothesis is about *which count is physical*, not *which domain* — the foundation doesn't move either way.

— Grace, 2026-07-27 [TEGMARK]. PRE-REGISTERED (before the count) Casey's "if 4, the subdomain forces 3": target-innocent IFF the fermion sub-locus is forced independent of + prior to the count. Forced candidates exist (chargeless S^4 locus, F571/F582/F583; SO(4,2) conformal EM sector, F66/Casey #14 — both independent of generations). Test: run threshold+signature on BOTH full tower and the forced sub-locus by the identical rule; sub-locus count is physical only if it's the forced carrier AND comes out by its own mechanism; if BOTH give 4, publish 4 (sub-locus can't override a genuine 4); E7 runs the identical sub-locus rule. Instinct sound + worth building; guard held (adopting it because it gives 3 = the trap). K955 (color forces the domain) stands either way.